

Lecture 4

In Class Exercises



Exercise 1

- Use the substitution method to show that the following recurrences have the solution specified:

$$T(n) = T(n - 1) + n \text{ has solution } T(n) = O(n^2)$$

Exercise 2

The solution to the recurrence $T(n) = 4T(n/2) + n$ turns out to be $T(n) = \Theta(n^2)$. Show that a substitution proof with the assumption $T(n) \leq cn^2$ fails. Then show how to subtract a lower-order term to make a substitution proof work.

Exercise 3

- Use the master method to give tight asymptotic bounds for the following recurrence:

$$T(n) = 2T(n/4) + 1$$